

3.3.15 Nuclear magnetic resonance spectroscopy (A-level only)

Chemists use a variety of techniques to deduce the structure of compounds. In this section, nuclear magnetic resonance spectroscopy is added to mass spectrometry and infrared spectroscopy as an analytical technique. The emphasis is on the use of analytical data to solve problems rather than on spectroscopic theory.

Content	Opportunities for skills development
Appreciation that scientists have developed a range of analytical techniques which together enable the structures of new compounds to be confirmed. Nuclear magnetic resonance (NMR) gives information about the position of ^{13}C or ^1H atoms in a molecule. ^{13}C NMR gives simpler spectra than ^1H NMR. The use of the δ scale for recording chemical shift. Chemical shift depends on the molecular environment. Integrated spectra indicate the relative numbers of ^1H atoms in different environments. ^1H NMR spectra are obtained using samples dissolved in deuterated solvents or CCl_4 The use of tetramethylsilane (TMS) as a standard. Students should be able to: • explain why TMS is a suitable substance to use as a standard • use ^1H NMR and ^{13}C NMR spectra and chemical shift data from the Chemistry Data Booklet to suggest possible structures or part structures for molecules • use integration data from ^1H NMR spectra to determine the relative numbers of equivalent protons in the molecule • use the $n+1$ rule to deduce the spin–spin splitting patterns of adjacent, non-equivalent protons, limited to doublet, triplet and quartet formation in aliphatic compounds.	Students should be able to use data in the Chemistry Data Booklet to suggest possible structures for molecules.